UI/API Automated Testing Process Management Rules
UI/API 自动化测试过程管理细则

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（仅供 BBA 内部使用）

Contents

1. Introduction 导言

1.1 Purpose 目的

Standardize and clarify the automated testing process, achieve unity of automated testing processes within the organization, improve overall testing efficiency, and provide guidance for automated testing activities.
规范并明确自动化测试流程，实现组织内自动化测试过程的统一，提升整体测试效率，并为自动化测试活动提供指导。

1.2 Definitions 术语定义

Automated testing: Automated testing is the process of transforming human driven testing behavior into machine execution. Usually, after designing test cases and passing the review, the testers execute the tests step by step according to the process described in the test cases, and obtain a comparison between the actual results and the expected results.
自动化测试：自动化测试是把以人为驱动的测试行为转化为机器执行的一种过程。通常，在设计了测试用例并通过评审之后，由测试人员根据测试用例中描述的过程一步步执行测试，得到实际结果与期望结果的比较。

2. Main roles and responsibilities 主要角色及职责

Roles Name 角色名称	Description 职责描述
Product owner 产品经理	• Provide corresponding requirement documents and output a list of key business scenarios . 提供对应的需求文档，输出关键业务场景列表。 • Cooperate in reviewing automated test cases and scenarios. 配合评审自动化测试用例和场景。 • Confirm automation test demonstration results. 确认自动化测试演示结果。
Test Management 测试管理岗	• Clarify the scope of automated testing. 明确自动化测试范围。 • Develop a testing plan. 制定测试计划。 • Requirement research, business scenario analysis, and confirmation of business scenario coverage. 需求调研、业务场景分析，确认业务场景覆盖程度 • Coordinate testing environment. 协调测试环境。 • Review test cases. 评审测试用例。 • Track the testing process, assist in solving problems during the testing process, and monitor risks during the testing process. 跟踪测试过程，协助解决测试过程中的问题，监控测试过

	程中的风险。 • Test report review. 测试报告审核。
Tester 测试执行岗	• Participate in testing requirement analysis. 参与测试需求分析。 • Design testing scenarios. 设计测试场景。 • Write test cases. 编写测试用例。 • Build an automated testing environment, develop and maintain automated testing scripts. 搭建自动化测试环境，开发并维护自动化测试脚本。 • Automated test case execution. 自动化测试用例执行。 • Analyze and submit defects, assist in defect localization. 分析并提交缺陷，协助缺陷定位。 • Output test results and complete test reports. 输出测试结果，完成测试报告。
Developer 开发人员	• Participate in determining the scope of automated testing. 参与自动化测试范围确定。 • Support the development of automated testing scripts. 支持自动化测试脚本开发。 • Defect repair. 缺陷修复。
Operations personnel 运维人员	• Test version release. 测试版本发布。 • Test environment management. 测试环境管理。

3. Preconditions 前置条件

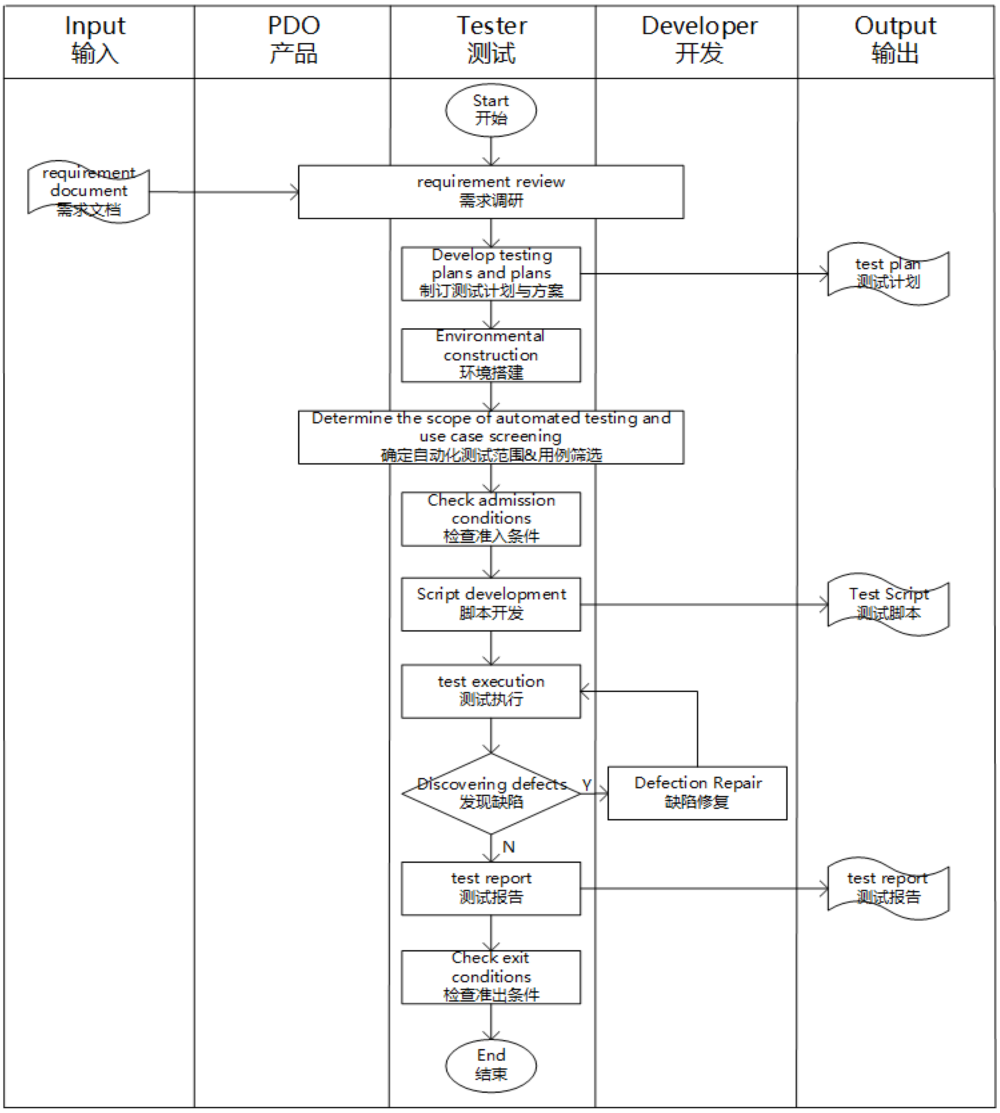
3.1 Entry Criteria 入口准则

Entry Criteria 入口准则
• The tested version has passed functional testing. 被测版本已通过功能测试。
• Stable automated testing environment. 稳定的自动化测试环境。
• The approved interface requirements document can be found on Jira/Confluence. 评审通过的接口需求文档，Jira/Confluence 上可查阅。

3.2 Input 输入

Input 输入
• Requirement documents that have been reviewed and approved. 评审通过的需求文档。
• Functional regression test cases. 功能回归测试用例。
• Business testing scenarios. 业务测试场景。
• Interface requirements document that has been reviewed and approved. 评审通过的接口需求文档。

4. Testing process 测试过程



4.1 Demand research and analysis 需求调研与分析

The automation testing management personnel conduct research on the business/interface requirements of the system to be tested based on the automation testing tasks.
自动化测试管理人员根据自动化测试任务，调研待测系统业务/接口需求。

4.1.1 Analyze the feasibility of project automation testing 分析项目自动化测试的可行性

Before conducting project automation testing, the first step is to confirm its feasibility and whether testing automation can be implemented. To carry out automated testing, the following prerequisites should be followed:
在进行项目自动化测试之前，第一步就是要确认其可行性，是否可以实行测试自动化。想要开展自动化测试，应该遵循以下几个前提条件：

- Frequent changes in software requirements
  软件需求变动不频繁
- The project cycle is long enough
  项目周期足够长
- Automated testing cannot immediately reduce the number of functional testing personnel
  自动化测试不能立马减少功能测试人员数量；
- Automated testing is not just about recording and replaying
  自动化测试不是简单的录制与回放；
- Automated testing is an assistance and supplement to manual testing, and cannot completely replace manual testing;
  自动化测试是对手工测试的辅助和补充，不能完全取代手工测试；
- Automated testing cannot cover all scenarios because there is not enough time or resources
  自动化测试不可能覆盖所有场景，因为没有足够的时间或资源；

4.1.2 Test requirement analysis 测试需求分析

In the testing requirements analysis stage, determining test coverage and automation test granularity, as well as screening test cases, are key tasks. The testing requirements are actually the testing objectives, and can also be seen as the functional points of automated testing. UI automation testing cannot achieve 100% coverage, only by maximizing test coverage. Testing requirements require the design of multiple automated test cases, and the analysis of testing requirements is used to determine the extent to which software automation testing needs to be achieved.
在测试需求分析阶段，确定测试覆盖率以及自动化测试颗粒度，以及测试用例的筛选等都是重点工作。测试需求其实就是测试目标，也可以看作是自动化测试的功能点。UI 自动化测试是做不到100%覆盖率的，只有尽可能提高测试覆盖率。测试需求需要设计多个自动化测试用例，通过测试需求分析判定软件自动化测试要做到什么程度。

UI automation testing is mainly considered from the perspective of actual business scenarios. Firstly, it needs to fit the positive scenarios used by users, and then filter out important reverse scenarios.

UI 自动化测试主要从实际的业务场景方面考虑，首先需要贴合用户使用的正向场景，然后再筛选出重要的逆向场景。

4.2 Develop testing plans and plans 制定测试计划与方案

The automation testing management position determines the testing scope, testing techniques and tools used, testing resource planning (including software and hardware resources), personnel arrangement, etc. based on the business requirements of the system to be tested, prepares automation testing plans, and plans the implementation of automation testing.
自动化测试管理岗根据待测系统的业务需求，确定测试范围、使用的测试技术及工具、测试资源规划（包括软硬件资源）、人员安排等，编制自动化测试方案，计划自动化测试实施。

Testing begins with the establishment of testing objectives and testing strategies, which should meet the requirements of the testing objectives. The testing plan includes evaluating the time required to complete all testing activities, arranging testing activities and allocating resources, controlling the testing process, and tracking the activities required to be taken throughout the entire testing process. These activities should be implemented before the project starts and run through the entire development process of the project. The testing plan is the most important activity in the testing process, including the following activities:
测试始于测试目标和测试策略的建立，测试策略应满足测试目标的要求。测试计划包括评估完成所有测试活动的时间，测试活动安排及资源分配，控制测试过程以及跟踪整个测试过程所需采取的活动，这些活动应该在项目开始前就实施，并贯穿项目的整个开发过程。测试计划是测试过程中最重要的活动，包括如下活动：

- Automated execution strategy: timed, build according to version, and manually triggered;
  自动化执行策略：定时、按照版本构建、人工触发；
- Schedule and deliver what results at what time
  进度安排，在什么时间交付什么成果；
- Personnel arrangement, based on the situation of team members, implement complex scripts with good skills; Capable of disassembling functional steps with strong business capabilities;
  人员安排，根据团队成员情况，技术好的复杂脚本实现；业务能力强的进行功能步骤拆解等；
- Risk assessment, estimating the risks during the project process.
  风险评估，对项目过程中的风险进行预估。

4.3 Determine the scope of automated testing 确定自动化测试范围

The automation testing management position determines the scope of automation testing based on the requirements document of the tested system (including interface requirements), functional test cases, relevant operation manuals, etc; Please ask the product manager and developers to confirm the scope of automated testing, focusing mainly on scenarios and use cases where automated testing cannot be conducted.
自动化测试管理岗根据被测系统需求文档（含接口需求）、功能测试用例、相关操作手册等，确定自动化测试范围；请产品经理和开发人员对自动化测试范围进行确认，主要关注不能进行自动化测试的场景和用例。

4.4 User Interface test case design/review UI 测试用例设计/评审

4.4.1 User Interface Automation Test Case Screening UI 自动化测试用例筛选

UI automation test cases can be used from functional test cases through filtering and simple modifications, with the main steps being:
UI 自动化测试用例完全可以从功能测试用例，通过筛选、简单修改使用，主要步骤为：

1) Filter functional test cases；
筛选功能测试用例；
2) Convert it into an automation use case template
将其转化为自动化用例模板；
3) Supplement or modify use cases that are not suitable for automation；
补充、修改不适用于自动化的用例；
4) Continuously maintain and optimize automation use cases.
持续维护和优化自动化用例。

Note: Based on the actual analysis of the product, some product functional test cases cannot meet the requirements for writing automation cases. Therefore, automation case writing can be carried out in conjunction with the original product requirements document.
注：根据产品实际情况分析，部分产品功能测试用例无法满足自动化用例的编写，可以采取结合产品原始需求文档进行自动化用例的编写。

The standard reference for screening functional test cases is as follows:
筛选功能测试用例的标准参考如下：

- Not all manual test cases need to be converted to automated test cases
不是所有的手工测试用例都要转为自动化测试用例；
- Considering the cost of script development, do not choose use cases with overly complex processes;
考虑到脚本开发的成本，不要选择流程太复杂的用例；
- The selected use cases should preferably be constructed into scenarios;
选择的用例最好可以构建成场景；
- The selected use cases can be repetitive and cumbersome parts;
选取的用例可以是重复执行，很繁琐的部分；
- The selected use case can be the main process, which is applicable to smoke testing.
选取的用例可以是主体流程，这部分适用于冒烟测试。

4.4.2 Scenario Review 场景评审

After the completion of the test case writing, for the completed test case scenario, members of the testing team will conduct an internal review first. After the internal review is approved, an external review meeting for the test case scenario will be organized.
测试用例编写完成后，针对编写完成的测试用例场景，测试组内成员先进行内部评审，内部评审通过后再组织测试用例场景外部评审会议。

4.5 Environmental construction 环境搭建

The automation testing implementation post can start building a testing environment in the early stages of use case design work. The establishment of a testing environment includes: the layout of the network environment, the installation and setup of testing tools, the invocation of testing hardware, and the deployment of the tested system.
自动化测试实施岗在用例设计工作开展的前期可着手搭建测试环境。测试环境搭建包括：网络环境的布置、测试工具的安装和设置、测试硬件的调用、被测系统部署等。

4.6 Script development 脚本开发

Automation test implementers develop automation test scripts based on the approved automation test cases or interface requirements, and according to the confirmed testing techniques and tools in the test plan. Unified management of automated test scripts, with automated test administrators organizing test implementers to conduct script reviews, ensuring compliance between script implementation and use cases.
自动化测试实施人员，根据已经评审通过的自动化测试用例或接口需求，按测试方案中已确认的测试技术及测试工具，开发自动化测试脚本。自动化测试脚本统一管理，自动化测试管理员组织测试实施人员进行脚本评审，脚本实现与用例的符合度。

4.6.1 Test script development 测试脚本开发

Record, debug, and write automation test scripts for each functional point based on requirement documents, automation test cases, and relevant operation manuals, and add checkpoints for parameterization. The output of this process is automated testing scripts and other common processing scripts for each functional point. The following should be noted during this process:
根据需求文档、自动化测试用例、相关操作手册，录制、调试、编写各个功能点的自动化测试脚本，并添加检查点，进行参数化。该过程的产出物是各个功能点的自动化测试脚本和其他公共处理脚本。此过程需要注意的是：

- The automation testing script code must be rigorous and standardized;
  自动化测试脚本代码必须严谨、规范；
- Automation testing scripts need to be developed based on automation test cases or interface requirements;
  自动化测试脚本需参照自动化测试用例或接口需求开发；
- Make automated testing scripts as intelligent, efficient, stable, and reusable as possible;
  尽一切可能使自动化测试脚本更智能、高效、稳定、复用性高；
- The development process often utilizes instrumentation and breakpoints to check for defects in business components or vulnerabilities in the code;
  开发过程多利用插桩+断点，检查业务组件是否存在缺陷或代码是否存在漏洞；
- After the script is developed and successfully run at least 3 times, it can be considered that the script has no problems.
  脚本开发完毕后，至少运行成功 3 次以上，方可认为脚本已经无问题。

4.7 test execution 测试执行

Automation testing implementers configure the test set and execution set according to the execution strategy in the test plan. Unified management of automated testing execution process, including configuring multiple task execution methods such as immediate execution, scheduled execution, Jenkins trigger execution, and configuring stakeholders for testing result notification. The debugging and execution of scripts require separate tasks to be established.
自动化测试实施人员，按测试方案中的执行策略，配置测试集和执行集。自动化测试执行过程统一管理，包括配置立即执行、定时执行、Jenkins 触发执行等多种任务执行方式，配置测试结果通知的干系人。脚本的调试和调试功能后的执行需要分别建立任务。

4.7.1 script execution 脚本执行

After the development of test scripts is completed, it is necessary to manage and execute them; The execution of the test script mainly includes the following content:
测试脚本开发完成后就要对测试脚本进行管理、执行；测试脚本的执行主要包含如下内容：

- Management configuration of testing environment
  测试环境的管理配置
- Test Script Configuration
  测试脚本配置
- Execution of test scripts
  测试脚本的执行
- Test exception interrupt handling and recovery
  测试异常中断处理和恢复

4.7.2 Analysis of test results 测试结果分析

Automation test results are generally divided into two situations - pass and failed. For test automation execution results, the main focus is on analyzing those results that did not pass:
自动化测试结果一般被分为 2 种情况——通过、失败。针对测试自动化执行结果，主要集中分析那些没有通过的结果：

- Is there a defect in the software product itself that causes the test execution to fail?
  分析是不是软件产品本身存在缺陷导致测试执行失败？
- Did a previous test execution fail, resulting in a large number of other test executions failing?
  是不是之前某个测试执行失败而导致其他大量的测试执行没通过？
- Check if the testing environment is abnormal?
  查看测试环境是否异常？
- If there are no issues with the product and testing environment, it is necessary to check the automated testing script. Is the testing script reporting false information? Is the verification point setting for the script incorrect?
  如果产品和测试环境都没有问题，就要检查自动化测试脚本，是不是测试脚本误报信息？还是脚本的验证点设置不对？

The case execution results and defects are first recorded on the automated testing platform. If any defects are found to be caused by the application, they will be imported into the defect management tool (Jira) for relevant developers to review and handle.
案例执行结果及缺陷先在自动化测试平台记录，如发现确系应用造成的缺陷，将导入缺陷管理工具(Jira)，由相关开发人员查阅、处理。

4.8 End of testing 测试结束

After the completion of automated testing, the system automatically notifies relevant stakeholders of the test results according to the configuration results.
自动化测试执行结束后，按照配置结果系统自动通知相关干系人测试结果。
The summary of automation testing includes the following content:
自动化测试总结包含以下内容：

- Summarize the testing progress of automated testing and the completion status of automated testing.
  总结自动化测试的测试进度，及自动化测试的完成情况。
- Review the testing process, summarize the problems encountered in automated testing and their solutions, in order to improve the skill level of testing personnel.
  回顾测试过程，总结自动化测试中遇到的问题及解决措施，以便提高测试人员的技能水平。

5. Post condition 后置条件

5.1 Exit criteria 出口准则

- The execution rate of automated testing cases reaches 100%;
  自动化测试的用例执行率达到 100%；

5.2 Output 输出

- Automated test scripts/use cases
  自动化测试脚本/用例
- Automation Test Report
  自动化测试报告